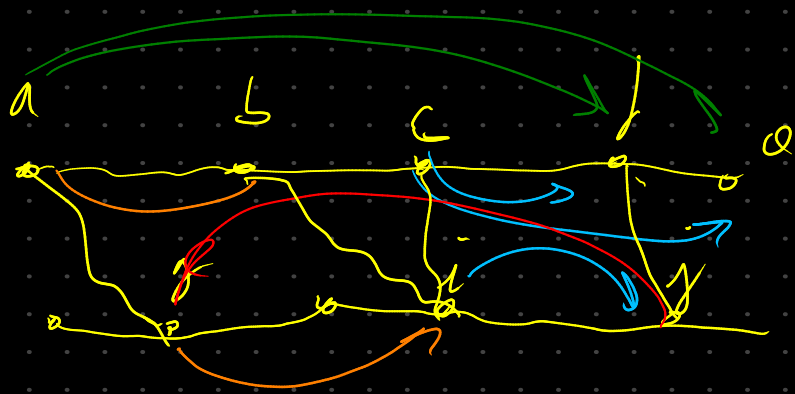
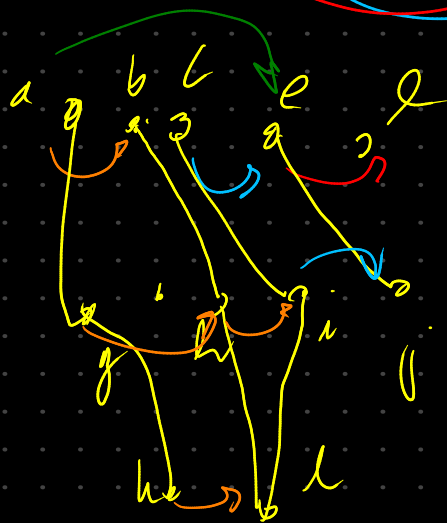
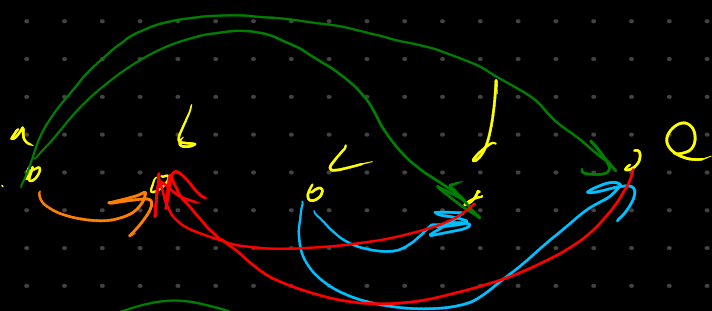


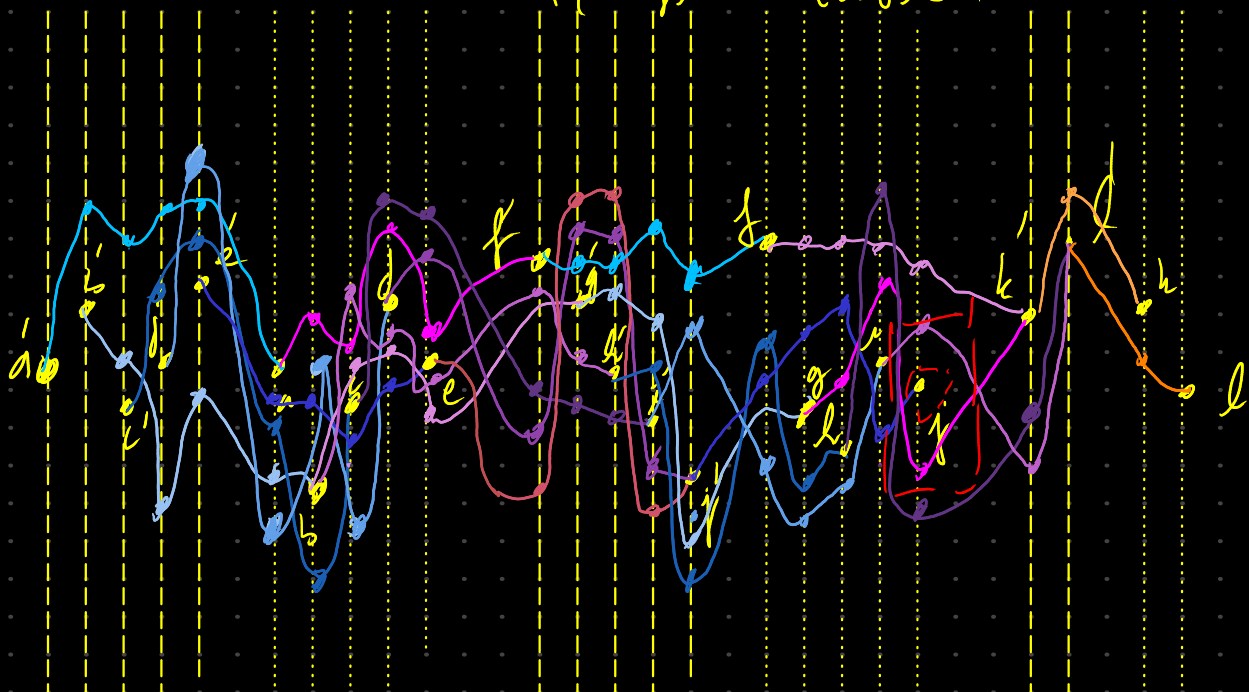


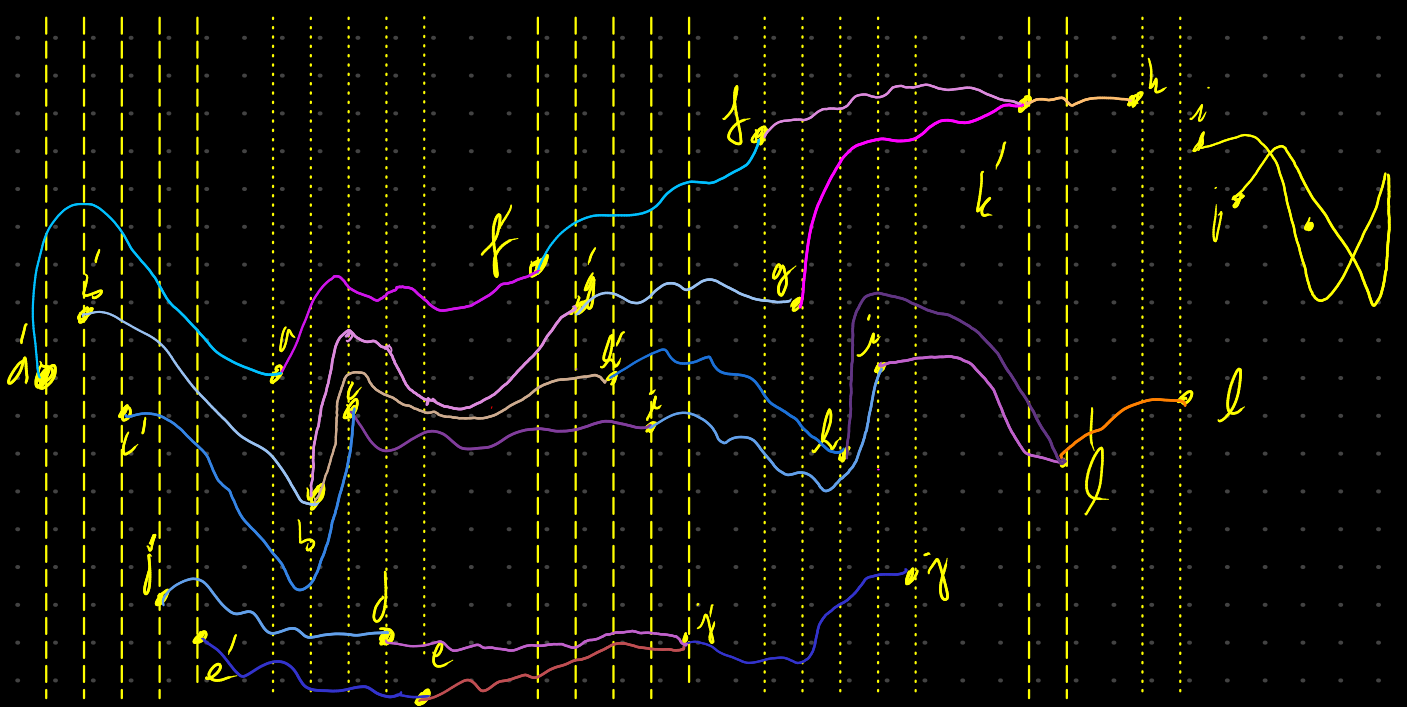
blue
green
red
handle



$$\begin{aligned}
 \varphi(ab) &= 1 & \varphi(ac) &= x & \varphi(ad) &= 1 & \varphi(ae) &= 1 \\
 \varphi(bc) &= x & \varphi(bd) &= 0 & \varphi(be) &= 0 \\
 \varphi(cd) &= 1 & \varphi(ce) &= 1 & \varphi(de) &= x \\
 \varphi(fg) &= x & \varphi(fh) &= 1 & \varphi(fi) &= x & \varphi(fj) &= 1 \\
 \varphi(gh) &= 1 & \varphi(gi) &= 1 & \varphi(gj) &= 0 & \varphi(hi) &= 1 \\
 \varphi(hj) &= 0 & \varphi(ij) &= 1
 \end{aligned}$$

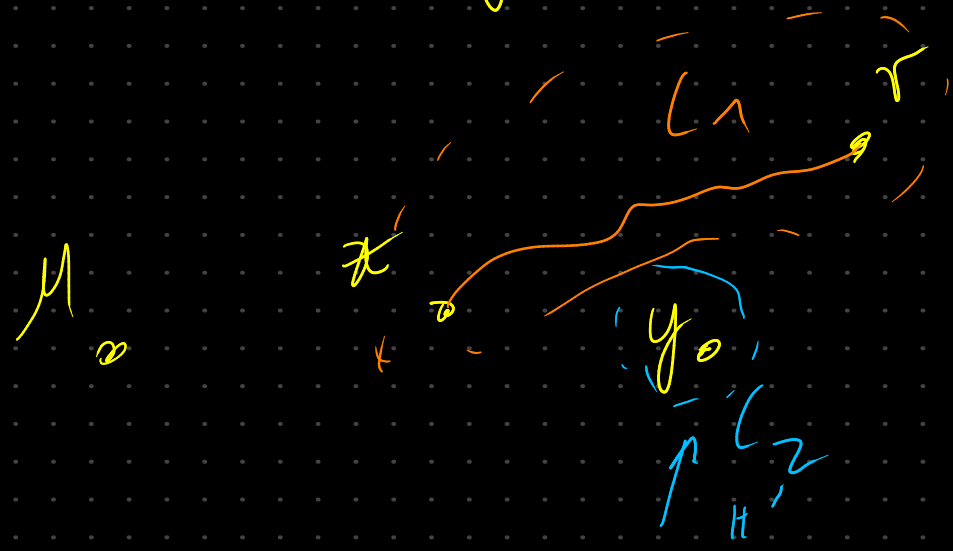
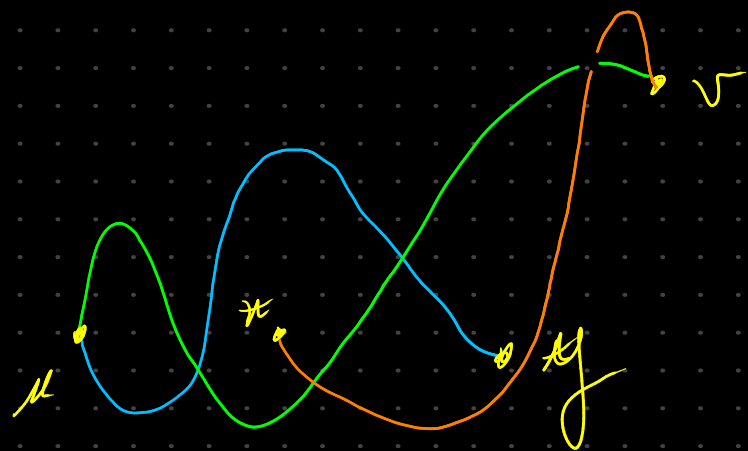
$$\begin{aligned}
 \varphi(hl) &= 1 \\
 \varphi(gh) &= 1
 \end{aligned}$$

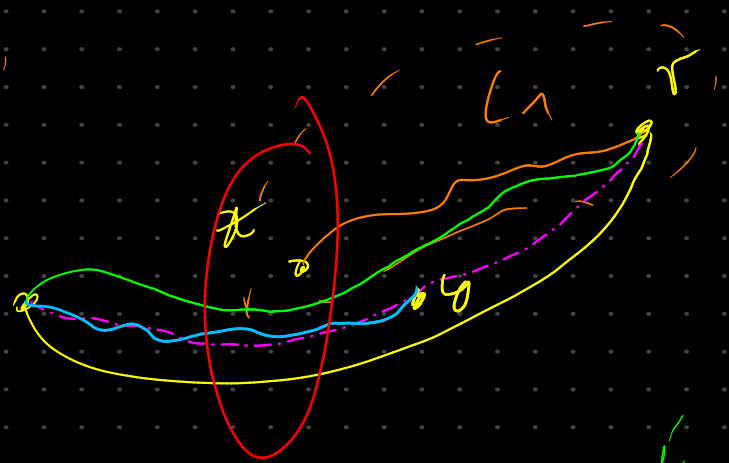
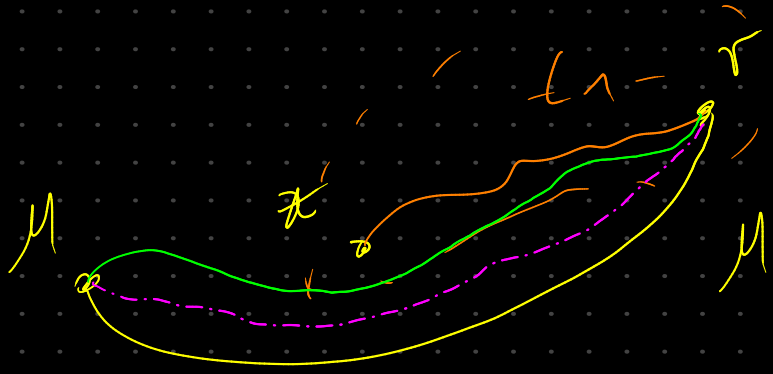
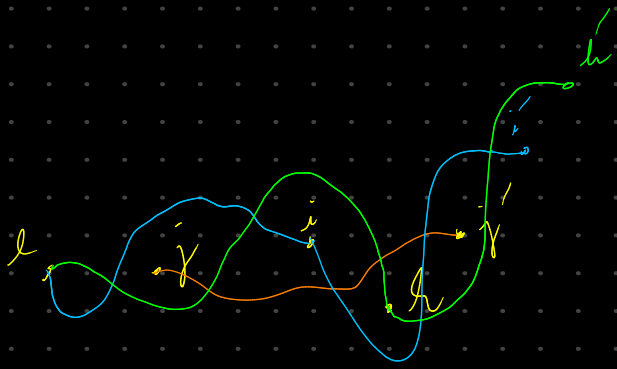




$$h \quad i \quad j$$

$$\varphi(hj) = 0 \quad \ell(ij) = 1$$





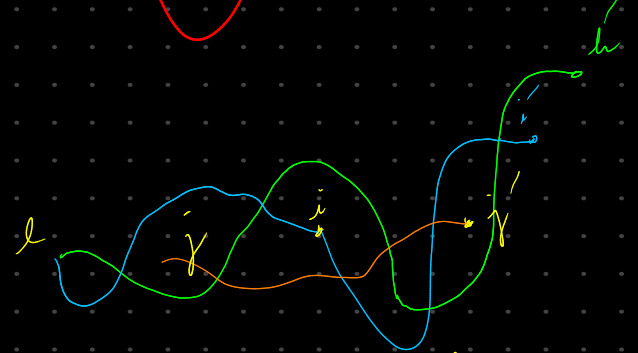
Conclusion

for a h disconnected graph

2SAT on every subgraph representing the connected comp.

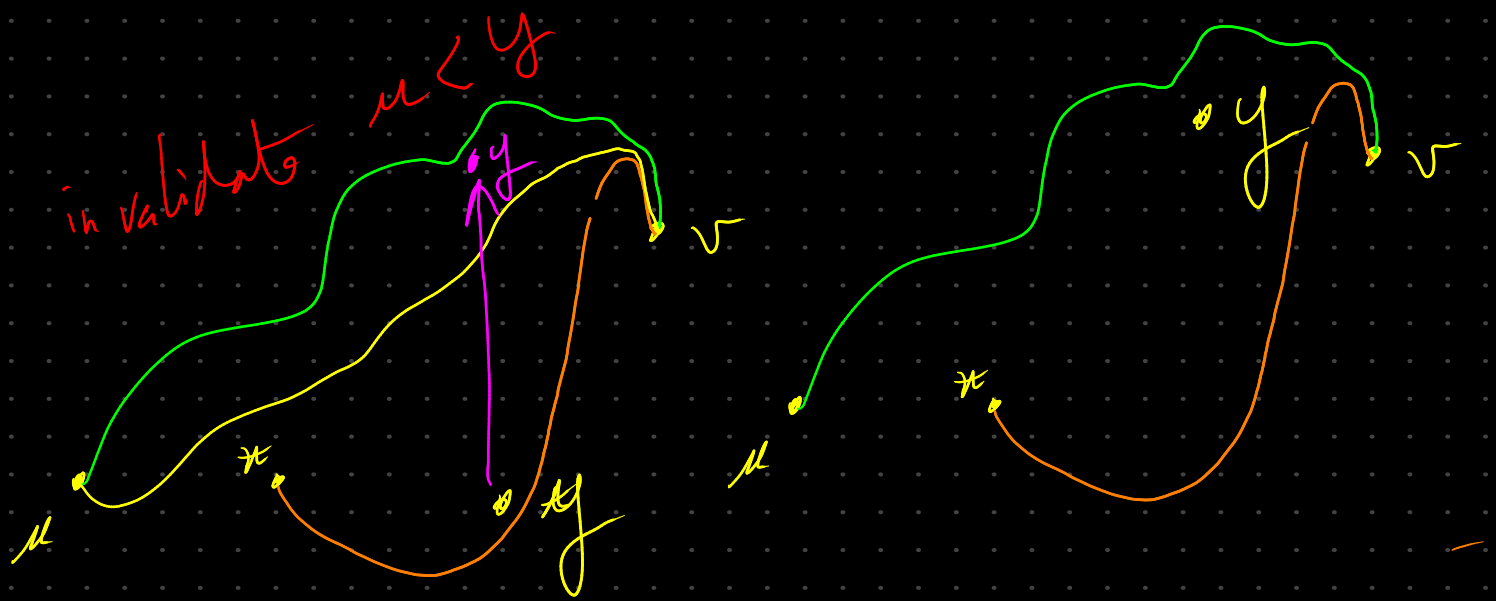
(we might find ourselves in the same case

if we want to keep the order



h connected

= if we find ourselves inside a cycle



Can I still redraw using Lemma 2.3
 even without leaving $\sim$ Cycle
 (we can add one) (for the
 case of
 invalidity)